

Jinglin (Ollie) Jian

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EDUCATION

Scripps Research

San Diego, California, US

Ph.D. Program (Kellogg Fellowship)

Aug 2025 – May 2029

Research focus: pretraining large models for biological representations, especially protein language modeling and dynamic 3D modeling

Ph.D. Advisors: Prof. Stefano Forli and Prof. Bryan Briney

University of Illinois Urbana-Champaign (UIUC)

Champaign, Illinois, US

M.S. in Information Sciences

Aug 2023 – May 2025

Coursework: ML for Bioinformatics, Text Information Systems, Large Language Models, Data Mining

Peking University (PKU)

Beijing, China

B.Econ. in Economics (Minor Program)

Sep 2021 – Jul 2023

Coursework: Statistics, Applied Econometrics, Causal Inference Models

Beijing Normal University (BNU)

Beijing, China

B.S. in Computer Science and Education

Sep 2017 – Jul 2021

Coursework: Data Structures, Databases, Web Development, OOP, Information Retrieval, AI

PUBLICATIONS

Google Scholar: <https://scholar.google.com/citations?hl=en&user=pXY2xYkAAAAJ>

[Ongoing] **Jian, J.**, Cai, Z., ..., Forli, S., & Briney, B. Compute-Optimal Scaling Laws for a Data-Constrained Protein Language Model. (2026).

[Under Review by NeurIPS] Lu, W., **Jian, J.**, ..., Wan, S., & Cong, L. From Protocols to Policies: A Long-Horizon Manipulation Benchmark from Scientific Workflows. (2026).

[Ongoing] Zhao, Z., ..., **Jian, J.**, ..., & Cong, L. Science Earth: Towards A Planet-Scale Operating System for AI-Native Scientific Discovery. *arXiv preprint arXiv:2606.01316* (2026).

[Ongoing] Torres-Paris, C., ..., **Jian, J.**, Holcomb, M., Forli, S., & Racki, L. R. Allosterically inhibiting *Pseudomonas aeruginosa*'s polyphosphate kinase 2A by disrupting its oligomerization. (2026).

[TechRxiv'25, Under Review by ACL] **Jian, J.**, Fung, Y. R., ..., Chen, Q., Lu, Z., & Wang, Q. Exploring Agentic Multimodal Large Language Models: A Survey for AIScientists. *TechRxiv* (2025).

[arXiv'24] Liu, H., Li, Y., **Jian, J.**, Cheng, Y., ..., & Wang, H. Toward a Team of AI-Made Scientists for Scientific Discovery from Gene Expression Data. *arXiv preprint arXiv:2402.12391* (2024).

[IEEE Big Data'24] **Jian, J.**, ..., & Chen, J. Big Data-Driven Computational Aptamer Design Framework via Parallel Monte Carlo Tree Search. *IEEE International Conference on Big Data* (2024).

[IEEE Big Data'24] Li, Z., **Jian, J.**, ..., & Zhang, Y. Patient Outcome Predictions via a Multimodal Language Model for Electronic Health Records. *IEEE International Conference on Big Data* (2024).

[Submitted] Hou, R., Zhou, D., **Jian, J.**, & Liu, G. GeoCM: Exploring Consistency Models and EGNs for Molecular 3D Structure Prediction. *UIUC Machine Learning for Bioinformatics Workshop* (2024).

[AIED'24] Xiao, Y., & **Jian, J.** Which Animal Would You Like to See on Your Flashcards? Designing Visual Aids Together with Kids Using GIMs. *The 25th International Conference on Artificial Intelligence in Education* (2024).

RESEARCH EXPERIENCE

[Ongoing] Data-Constrained Scaling Laws for Protein Language Modeling

Jul 2026 – Present

Advisor: Prof. Bryan Briney (Scripps)

- Developed a $\mu\text{P}/\text{CompleteP}$ training pipeline for learning-rate transfer across 20M–800M-parameter models and built iso-FLOP ladders to fit (A, α) and estimate $N^*(C)$.
- Developed fresh-data and repetition experiment families to fit (B, β, E) and R_d^* , yielding a scaling law that predicts model performance under the given compute budgets and a finite protein sequence dataset; validated the law at held-out compute budgets.
- Extending the framework to compare masked language model, autoregressive, and diffusion-based protein LMs at equal pretraining FLOPs.

[Ongoing] Antibodies for All

May 2026 – Present

Advisor: Prof. Bryan Briney (Scripps)

- Developing data-constrained scaling laws that characterize how model size, unique antibody sequences, and data repetition jointly determine antibody language model performance.

[Ongoing] Dark Proteome: Molecular Dynamics of All Human Proteins

May 2026 – Present

Advisor: Prof. Stefano Forli (Scripps)

- Generating molecular dynamics simulations across $\sim 20,000$ human proteins, producing conformational ensembles, electrostatic maps, and nucleophilicity scores.
- Developing large models to learn molecular representations from proteome-wide simulations for downstream tasks, including identifying covalent drug-discovery hotspots missed by static structures.

[Ongoing] LabOS²: From Dry-Lab Planning to Wet-Lab Execution

Mar 2026 – Present

Advisors: Prof. Le Cong (Stanford), Prof. Mengdi Wang (Princeton)

- Designed protocol-inspired robotics tasks by mining 5,628 articles to identify 2,877 executable workflows across four evaluation levels (L1–L4) with scene randomization; the benchmark is under review by NeurIPS 2026.
- Designed and developed a pipeline to generate 3D laboratory assets with visual meshes and physical collision models, and finalized a multi-level, multi-asset, multi-scene simulation dataset on Kaggle.
- Benchmarked recent vision-language-action models, including $\pi_{0.5}$, ACT, SmolVLA, and X-VLA, finding a 21% success rate on single-primitive tasks and 0% on full protocols across L1–L4.
- Developing an integrated robotic system that connects protocol planning, perception, manipulation, and execution to enable end-to-end cell-culture workflows.

[Ongoing] Protocol: AutoDock6 for High-Throughput Virtual Screening

Dec 2025 – Present

Advisor: Prof. Stefano Forli (Scripps)

- Preparing AutoDock6 scripts and practical tutorials for high-throughput virtual screening on computing clusters.
- Developing agent skills and worked examples to streamline end-to-end screening workflows.

Virtual Screening Pipeline for Antibiotic Discovery

Nov 2025 – Jan 2026

Advisors: Prof. Lisa Racki (Scripps), Prof. Stefano Forli (Scripps)

- Established a virtual screening pipeline targeting Ppk2A in *P. aeruginosa*, a WHO critical-priority pathogen; used **AutoDock-GPU** and **AlphaFold3** to screen **2.4 million** compounds and identified **23 candidates** now undergoing wet-lab validation.

[TechRxiv'25] Exploring Agentic Multimodal Large Language Models: A Survey for AIScientists *Jun 2025 – Nov 2025*

Advisors: Profs. Qingyun Wang (William & Mary), Qingyu Chen (Yale), and Zhiyong Lu (NIH)

- Led a survey framing AI scientists through machine learning and agentic multimodal LLMs.

[Submitted] GeoCM: Exploring Consistency Models and EGNNs for Molecular 3D Structure Prediction *Oct 2024 – Dec 2024*

Advisor: Prof. Ge Liu (UIUC)

- Combined **Equivariant Graph Neural Networks (EGNNs)** with **Consistency Models** (a diffusion-based framework for one- or few-step sampling) for self-supervised molecular 3D structure prediction.
- Defined **Coverage Rate** and **Matching Error** to benchmark GeoCM against baselines, demonstrating state-of-the-art performance.

[Big Data'24] Fast and Accurate Drug Discovery Framework *Jan 2024 – Oct 2024*

Advisors: Prof. Yang Zhang (UIUC) and Dr. Jin Chen (Cleveland Clinic)

- Developed an enhanced parallel **Monte Carlo Tree Search** framework for aptamer design, achieving a **98-fold** improvement in computational efficiency and a **7.59-fold** improvement in sequence quality.

[arXiv'24] Team of AI-made Scientists (TAIS) *Aug 2023 – Feb 2024*

Advisor: Prof. Haohan Wang (UIUC)

- Introduced **TAIS**, an LLM-based framework that automates ML workflows for disease-predictive gene discovery from expression data, outperforming **GPT-4**, **MetaGPT**, and **AutoGPT**.
- Retrieved data from **GEO** and **TAGC** and created **GenoTEX**, a benchmark for genomic-data analysis with gene-symbol alignment and statistical corrections.
- Built **Data Engineer**, **Code Reviewer**, and **Domain Expert** agents with template-based code generation, subprocess execution, structured logging, and inter-agent communication.

[Bachelor's Thesis] Semi-Automatic Knowledge Graph Construction *Sep 2020 – Jul 2021*

Advisor: Prof. Qinhua Zheng (BNU)

- Developed a **semi-automatic knowledge-graph construction** pipeline that combines **supervised learning** with **human feedback** to reduce manual annotation time.
- Developed a benchmark dataset for educational entities by annotating transcriptions using BIO tagging.
- Refined a supervised **BiLSTM-CRF** model for **entity recognition** using dynamic term re-ranking based on **mutual information** and **human feedback**, improving the F1 score from 0.54 to **0.76**.

PUBLIC SERVICE

Organizer, Claw4S Conference; Co-Creator, clawRxiv *2026 – Present*

- Co-organized the inaugural agent-native conference for executable scientific skills and co-created clawRxiv, attracting **490 AI agents** and **2,867 papers** in its first year.

Core Contributor, LabClaw (1K+ stars; open-source skill stack for biomedical AI agents) *2026 – Present*

- Contributed to a library of **240 production-ready SKILL.md files** for biomedical AI workflows spanning biology, lab automation, vision/XR, drug discovery, medicine, data science, literature research, and scientific visualization.

Reviewer, Conference on Neural Information Processing Systems (NeurIPS) *2026*

Reviewer, IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) *2026*

PROFESSIONAL EXPERIENCE

Summer Software Engineer Intern

May 2024 – Aug 2024

Supervisors: David Bachtler and Ian Cowen, ReDirect

- Engineered subscription functionality using the Flutter framework and implemented unit tests.

Research Assistant

Sep 2023 – Aug 2024

Supervisor: Prof. Mackenzie Alston, University of Illinois Urbana-Champaign

- Conducted a literature review of randomized controlled trials (RCT) using Zotero and collected more than 2,000 email addresses through web scraping.

Teaching Assistant, IS 589 Independent Study

Aug 2023 – Dec 2023

Supervisor: Prof. Haohan Wang, University of Illinois Urbana-Champaign

- Developed and published a comprehensive machine-learning tutorial, the “Gold Standard Pipeline Guide,” teaching students to analyze gene-expression data for disease-related gene discovery. [[GitHub](#)]
- The resource has been forked by nine students for use in their own research projects.

Head of the Online Learning Department & Volunteer Teacher

May 2019 – Aug 2022

China Starry Night (NGO) [[Web](#)]

- Produced online educational videos that helped rural artisans preserve intangible cultural heritage by documenting and sharing traditional crafts with younger generations.
- Published courses on Bilibili, China’s leading video platform, garnering more than 10,000 views.
- Presented the project at the 5th China Education Innovation Expo (National Award; top 1%).

HONORS AND AWARDS

- **Kellogg Fellowship**, Scripps Research; Kellogg Family and the ALSAM Foundation Aug 2025 – Aug 2028
- **National Innovation and Entrepreneurship Training Award (1%)**, Ministry of Education 2021
- **Outstanding Teaching Practice (1%)**, Beijing Normal University 2021
- **Jianghaiziqiang Fellowship (1%)**, Beijing Normal University 2020
- **First-Class Scholarship for Competition Excellence (1%)**, Beijing Normal University 2019
- **First-Class Scholarship for Academic Excellence (10%)**, Beijing Normal University 2017 – 2021

SKILL SET

Machine Learning & NLP	TensorFlow, PyTorch, LangGraph, LangChain, scikit-learn, NLTK
Programming	Python, Java, C, JavaScript, Stata
Web & Databases	React, Node.js, RESTful APIs, HTML, MySQL, MongoDB, Neo4j
Cloud	AWS S3, EC2, API Gateway
Development Tools	Git/GitHub, Docker, unit testing, L ^A T _E X, Tableau